DIVYANSH RAJAT



Computer Science Engineering Student | Software Engineer | AI/ML Enthusiast

Email: divyanshrajat1999@gmail.com | **Mobile:** +91 9315324721
linkedin.com/in/divyansh-rajat-4857a01a4 | github.com/divyanshrajat
Punjab, India

PROFESSIONAL SUMMARY

Computer Science undergraduate with a strong foundation in **Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks,** and **Object-Oriented Programming**, with hands-on experience in **Python, full-stack development, and data-driven applications**. Actively builds practical projects and explores **Artificial Intelligence, Machine Learning, and data analysis**, with two research papers presented at national and international AI/ML conferences. Seeking internship and research opportunities with technology organizations and academic mentors to contribute to **AI/ML systems, backend development, and scalable software solutions**.

EDUCATION

Degree / Qualification	Institution	Year	CGPA / Score
B.Tech in Computer Science & Engineering (Pursuing)	BBSBEC, I.K. Gujral Punjab Technical University, Punjab, India	2023 – 2027	CGPA: 7.9 (till 6th Sem, 1 Backlog)
Senior Secondary (CBSE, Class XII, Non-Medical)	Central Board of Secondary Education, Delhi, India	2017	59.9%
Matriculation (CBSE, Class X)	Central Board of Secondary Education, Delhi, India	2015	75%

KEY PROJECTS

PRAMAAN – Secure Digital Testimony & Evidence Platform ***May 2026***
GitHub: github.com/asterm214-hue/PRAMAAN

- Built a full-stack, trauma-informed web platform for secure collection, organization, and management of survivor testimonies and digital evidence via text, audio, and multimedia workflows.
- Designed responsive, accessible frontend interfaces using **React, TypeScript, Tailwind CSS, and Framer Motion**.
- Engineered backend APIs with **Python and FastAPI**, implementing **JWT-based authentication**, input validation, and protected data workflows.
- Built timeline-based evidence organization and case documentation features; collaborated in a hackathon team on architecture, UI/UX, and system integration.

Tech Stack: React, TypeScript, Vite, Tailwind CSS, FastAPI, Python, SQLAlchemy, SQLite, JWT

AI-Based Plagiarism Detection System ***Jan 2026 – May 2026***
GitHub: github.com/divyanshrajat/AI_Plagiarism_Detection_System

- Built a **Flask** web application for plagiarism detection using **NLP and semantic similarity techniques**.
- Implemented **Sentence-BERT embeddings and cosine similarity** to detect paraphrased/semantically similar content, plus an AI-generated text detection module.
- Designed an end-to-end pipeline for text extraction (PDF, DOCX, TXT), preprocessing, and analysis, with a REST backend and role-based auth (Student/Teacher/Admin) in **MongoDB**.
- Generated structured plagiarism reports with similarity scores for academic evaluation.

Tech Stack: Python, Flask, NLP (NLTK, Sentence Transformers), MongoDB, HTML, CSS

Secure Chat Application (Socket Programming + RSA) ***Jun 2025 – Jul 2025***
GitHub: github.com/divyanshrajat/Chat-application

- Developed a client-server chat application using **Python socket programming** for real-time communication.
- Implemented **RSA-based asymmetric encryption** for secure message transmission across a multi-client concurrent system using threading.

Tech Stack: Python, Socket Programming, Cryptography (RSA)

RESEARCH & PUBLICATIONS

Natural Language Processing in Software Applications: A Comprehensive Review

- Published in an ISBN-edited book (ISBN: 978-81-966131-4-3); presented at the 37th National Convention of Computer Engineers & AIML-2024 Seminar.
- Analyzed NLP techniques including **BERT, GPT models, sentiment analysis, chatbots, and machine translation**, covering applications, challenges, and future trends.

\Ethical and Responsible AI: A Comprehensive Review of Principles, Methods, and Tools

- Presented at the International Conference on Innovations and Applications of AI (ICIAI 2025), organized with **IEEE Chandigarh Section** and supported by DST, Govt. of India.
- Published in the International Journal of Information Security Engineering (IJISE), Vol. 4, Issue 1 (2026); explored AI ethics, fairness, explainability, and privacy-preserving techniques (**SHAP, LIME, Federated Learning, Differential Privacy**).

SKILLS & TECHNICAL COMPETENCIES

Programming Languages: Python, C++, SQL

Computer Science Fundamentals: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Computer Architecture

Software Engineering: OOP, Linux, Shell Scripting, Git, Debugging, Testing, Design Patterns, Multithreading

Backend & APIs: FastAPI, REST APIs, Authentication (JWT), Multithreading

Databases: MySQL

AI / Machine Learning: NumPy, Pandas, Scikit-learn, PyTorch

Security: Secure Coding, API Security, OWASP

Math Foundations: Linear Algebra, Probability, Statistics, Calculus, Optimization

Developer Tools : VS Code, Jupyter Notebook, Google Colab

ACHIEVEMENTS & HACKATHONS

- Secured **3rd Position** at Saviskar 2025 Tech Event, CGC University, Mohali *(Oct 2025)*
- Participated in **National Level Ideathon (Junk2Genius 2025)** – RIMT University, Mandi Gobindgarh, Punjab
- Participated in **Quantum 3.0 Vibe Coding Hackathon**, GGI, Khanna, Punjab *(Feb 2026)*
- Participated in **Hack-N-Win 3.0 Hackathon**, CGC University, Mohali *(Mar 2026)*

LEADERSHIP & RESPONSIBILITY

Lead Coordinator, College Tech Fest ***Apr 2026***

- Managed event planning, team coordination, and technical execution.

Organizer, Alumni Meet ***Nov 2025***

- Handled registration workflow and attendee management for a large-scale event.
- Designed event mementos, contributing to branding and presentation; collaborated with cross-functional teams for seamless execution.

CERTIFICATIONS

Machine Learning & Python Workshop – NIELIT Chandigarh ***Mar 2025***

- Covered Python, NumPy, Pandas, Supervised & Unsupervised Learning.

Cyber Security Certification – NIELIT Delhi ***Jun 2025 – Jul 2025***

- Topics: Network Security, Cryptography, OSINT, Vulnerability Assessment.